AI ACADEMY · LEVEL 4 · GRADE 9 · WEEK 5 · TEACHER GUIDE

Inspect Before Modeling

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

THE CONNECTED WEEK

Teach inspection as documented inquiry.

Use row-level flags, explicit denominators and separate keys for the optional unit extension.

Keep one habit: Reserve the fresh export and require bounded conclusions even when a card looks complete.

MAIN LESSON · SECTION 1

Prepare the connected experiment

> Keep prior exposure, shared practice and independent evidence distinct.

Goal and main teaching plan

Primary target: Inspect row meaning **missing value**s and repeated records

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Materials and prerequisite

Teach inspection before modelling in two 45-minute sessions. Use fictional data only. Emphasize the unit of **observation**, **provenance**, units, unknown values and overlapping audit flags.

The **dataset card** is a concise classroom documentation exercise inspired by datasheets for datasets, not a claim to implement every part of a research framework.

Students must distinguish **observation**s from unsupported repairs and from **evidence** of model fitness.

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

It is a numerical quantity whose differences measure time. Whole-number formatting does not turn it into a category.

Printed lesson and workbook

Cards for the five raw rows including both B rows

Optional Python 3 editor; no packages required

Blank card headings: purpose source row **schema** findings changes limits version owner

Apply the connected idea: Units conversions and data provenance

Teach Week 15 as a two-session investigation using fictional data. Two fictional activity logs use different duration units. Log A records 180 seconds and Log B records 4 minutes.

Keep the **raw value**s, units and source labels; derive a shared unit so another learner can reproduce the comparison. Assess the three stated outcomes through the independent task, not by copying the worked result.

Readiness answer

3 minutes, since 60 seconds make one minute.

Python 3 editor with a fresh local practice file

Paper trace table and the student workbook

Fictional **input** records from this lesson; no real student identifiers

Opening retrieval

Without opening the old lesson, explain one key idea from Python and Table Readiness. Give an example and a mistake that the idea helps you avoid.

Look for: Specify a table **schema**

Reference example, not the only acceptable wording: activity: nonempty text; minutes: integer from 0 to 60 inclusive.

Without opening the old lesson, explain one key idea from Features, Labels, Targets. Give an example and a mistake that the idea helps you avoid.

Look for: Separate **prediction input**s from observed targets

Reference example, not the only acceptable wording: Candidate features are pages and queue_at_checkout.

The target is returned_late and the tracking field is loan_id. days_to_return is known only after return and directly defines the target, so it must not enter checkout **prediction**s.

End goal

By the end, I can explain inspect before modeling using a traced example and a stated limitation.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Case exposure check: this source example appears in ai-4/5, ai-4/7, ai-4/9. Reuse a familiar case for recall or modelling; verify that the case used to assess transfer has not already received feedback.

Retrieval answer guidance

Python and Table Readiness

Features, Labels, Targets

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

A table contains 8 cm, -2 cm, missing and two copies of the same leaf photo. Propose a separate inspection for each issue.

Concept to explain

Dataset **shape**, types, **range**s, **duplicate**s, and **missingness** determine whether modeling results can be trusted.

Term	Meaning
shape	number of rows and columns
range	smallest to largest plausible value
duplicate	repeated example or record
missingness	where values are absent

Prepare the class

This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Readiness answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach issue counts with explicit overlap

> Keep the trip export and separate height picture labelled throughout.

Source, row contract and complete raw table

A dataset can look like a neat table and still contain missing outcomes, repeated records or impossible values. Before **fitting** a model, inspect what one row represents and document where the table came from.

A **dataset card** is a short record of its purpose, contents, collection, changes and limits. It helps another person decide whether the data suits a proposed use.

Our source is a teacher-authored fictional export called Robot practice v1, created for this lesson. Each row is intended to describe one trip, with one unique trip ID and elapsed duration in minutes.

Durations may be zero or positive; None means unknown. All trips are described as indoor practice trips. No real robot measurements or personal records were collected, and there is no **evidence** about outdoor performance.

Trip ID	Duration in minutes	What is visible
A	8	A recorded duration
B	None	An unknown duration
B	None	The same ID appears again
C	-2	A negative duration
D	12	A recorded duration

Keep this raw version unchanged. At this stage a flag means investigate, not automatically delete or replace. A repeated ID might be a copied row or a problem with the supposed one-row-per-trip definition.

We need **evidence** before deciding what happened.

Trace the four output counts and limits

A set stores each distinct ID once. The **output** is 5 4 2 1: five rows, four distinct IDs, two rows with unknown duration and one row with a negative duration. There is one extra occurrence beyond the four distinct IDs.

This is not a count of all rows in repeated-ID groups: both B rows belong to that group.

The code assumes every row is a pair and each duration is a number or None. It is a small audit, not full **schema** validation. It does not check text values, units, source truth or whether IDs refer to real unique events.

A successful run does not certify data quality.

Term	Meaning
Provenance	Where data came from and how it was created
Schema	The stated meaning types units and rules for fields
Missing value	An unknown or unavailable value rather than a measured zero
Dataset card	A concise record of dataset contents creation intended use and limitations

Draft the card without certification claims

For Robot practice v1, record the purpose as practising inspection, the source as teacher-authored fictional data, and the row meaning as an intended single indoor trip.

Record the five-row count, the ID field, minutes as the duration unit and None as unknown. List the **duplicate**-ID and invalid-duration findings.

State that no cleaning has occurred, no train/test split is assigned, and no model has been evaluated.

Add intended use, excluded uses, known gaps, version and a contact or owner. Here the lesson teacher maintains the practice copy; its allowed classroom purpose is inspection practice.

Do not describe it as validated navigation data.

For a real external dataset, also verify its actual permission or licence and collection context; an absent field in the documentation is an unanswered question, not permission to invent a value.

> **What fraction of the five raw rows has unknown duration? Can you subtract two missing rows, one negative row and one extra ID occurrence from five to count distinct usable trips? Explain using the B rows.**

Read the pictured heights

Spot a **duplicate**, an impossible value and a valid boundary before **fitting**.

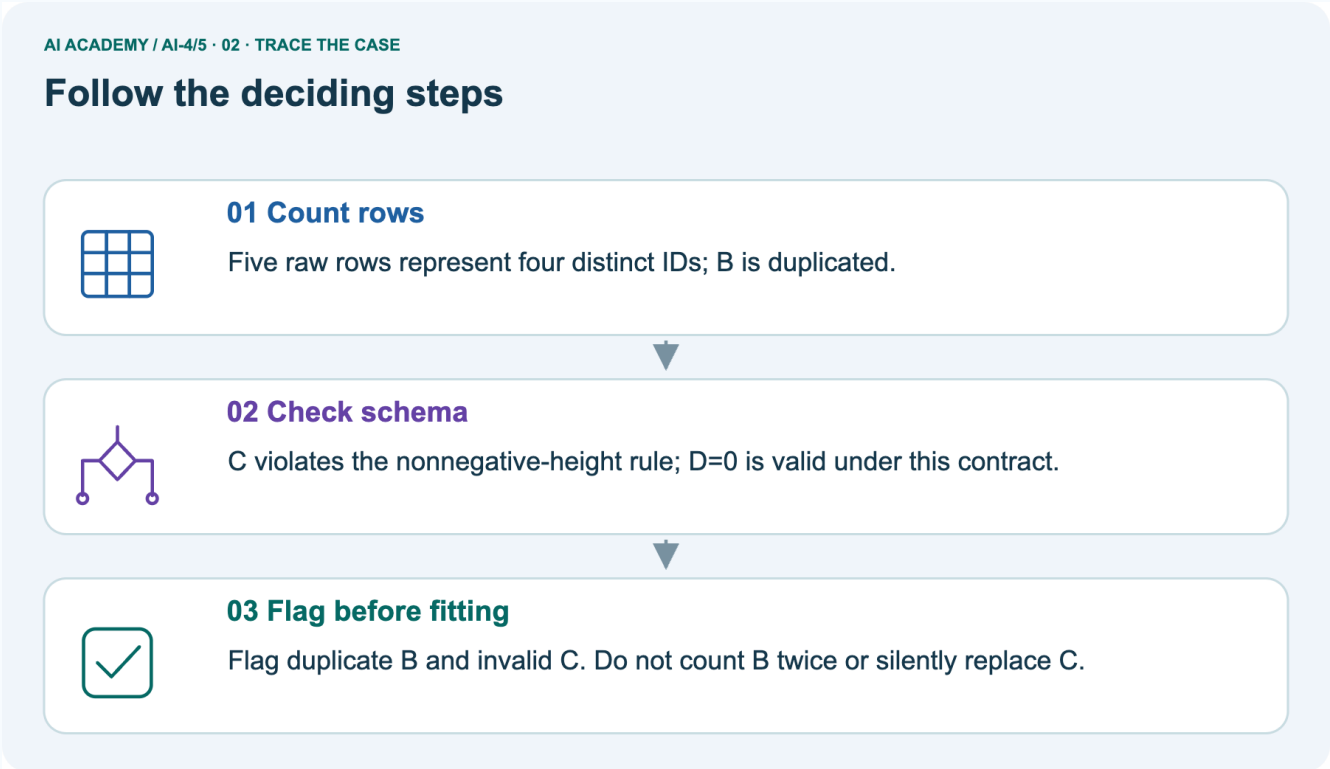
Toy heights table: A=3, B=5, B=5, C=-2, D=0 cm. Each plant should have exactly one nonnegative height record.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Keep the examples separate. Robot practice v1 has five raw rows, four distinct IDs, two missing-duration rows and one negative row; the extra B occurrence overlaps missingness. The pictured height table instead has A=3,B=5,B=5,C=-2,D=0 cm and no missing values. Its duplicate and invalid-value issues are separate.

A confirmed accidental copy can be removed only in a documented new version; resolving B does not repair C. Zero is allowed by both stated contracts, although validity does not establish measurement truth.

Trace separate duplicate and invalid-value flags



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Count rows	Five raw rows represent four distinct IDs; B is duplicated.
2. Check schema	C violates the nonnegative-height rule; D=0 is valid under this contract.
3. Flag before fitting	Flag duplicate B and invalid C. Do not count B twice or silently replace C.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Read the picture in order. Flag **duplicate** B and invalid C. Do not count B twice or silently replace C. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

A plausible nonnegative value can still be incorrectly measured; **schema** checks do not establish truth.

Predict an evidenced duplicate removal

Change: The teacher confirms B's second row is an accidental **duplicate**.

Prediction: Remove that **duplicate** with a recorded reason; C still needs correction or explicit exclusion.

Reason: Resolving one defect does not resolve a separate impossible measurement.

▼ Optional unit/provenance explanation

Two fictional activity logs use different duration units. Log A records 180 seconds and Log B records 4 minutes.

Keep the **raw value**s, units and source labels; derive a shared unit so another learner can reproduce the comparison.

The script prints [3.0, 4]. The slash operator returns a floating-point result. The original dictionaries are unchanged. Assume nonnegative numeric **raw value**s have already been validated.

An unsupported unit produces None here and must be reported for review rather than included in arithmetic.

1.5 minutes and 0.0 minutes. Two minutes is longer. Exit: No. Mark it unresolved and ask for the unit definition. Do not include it as a known duration.

▼ Earlier wording, alternatives and picture transcripts

Explain the mechanism

Trace the information, fitted behavior, and evaluation proof separately.

QUESTION Produce a data-quality report before **fitting** any model.

MODEL IDEA Dataset **shape**, types, **range**s, **duplicate**s, and **missingness** determine whether modeling results can be trusted.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

UNDER THE HOOD Dataset **shape** is 40 rows × 5 columns. Inspection finds 3 **duplicate** rows and 4 missing light values.

Draw a fresh data → model → **prediction** → evaluation trace:

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

STEP 1 Dataset **shape** is 40 rows × 5 columns.

STEP 2 Inspection finds 3 **duplicate** rows and 4 missing light values.

STEP 3 Leaf widths **range** from 0.4 to 48 cm; inspect the extreme value and unit.

STEP 4 Document, repair, or exclude by a stated rule—never silently delete inconvenient rows.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

Concrete teaching case

Lengths include 5 cm, blank and -3 cm, plus a **duplicate** photo row.

Name checks and a deletion risk.

Exact explanation to model

Check **missingness**, physical validity, units and **duplicate** identity. Investigate and document repair. Blind deletion can disproportionately remove a relevant condition.

Explain the added worked example

Key terms

Apply the connected idea: Units conversions and data provenance

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

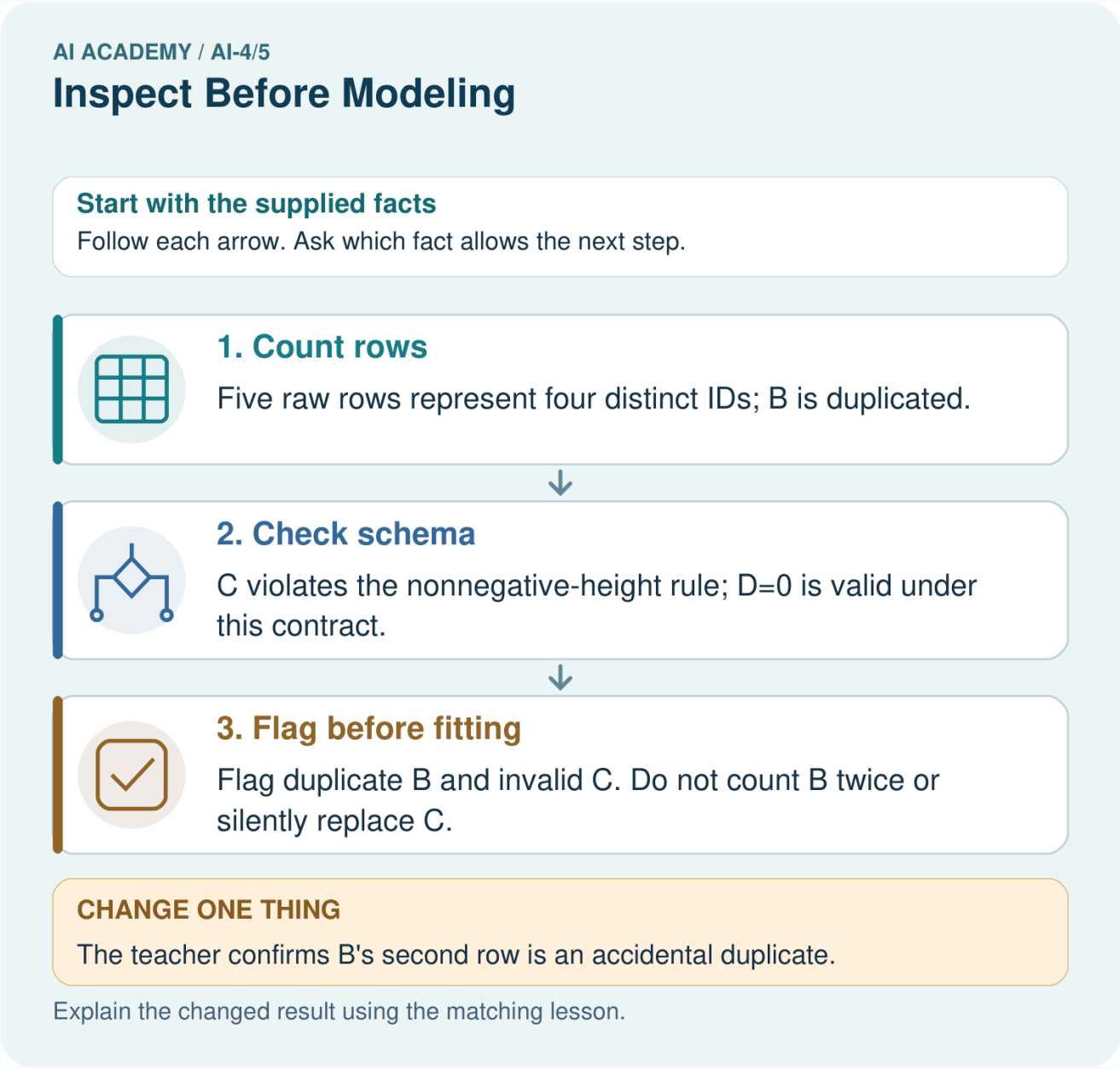
These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Inspect Before Modeling

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Guide the calculation, then remove help

> Collect each learner's prediction before showing its key.

Current completion task and checked key

Completion task: Explain each number in 5 4 2 1. How many extra ID occurrences are there, and how many rows belong to a repeated-ID group?

Teacher check: The numbers mean 5 raw rows, 4 distinct IDs, 2 missing durations and 1 negative duration. There is 1 extra ID occurrence; 2 rows belong to the repeated-ID group B. These are different counts.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for Dataset inspection and cards. Proposed explanation: "A complete card certifies data quality". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: The card records **evidence** and gaps; its claims still need checking. Documentation does not prove suitability.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial question and changed-case reasoning

Which original information should the inspection log preserve? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Keep raw row count 5, distinct IDs 4, **duplicate** B and invalid C, plus any confirmed correction and its reason. Changed-case check: Remove that **duplicate** with a recorded reason; C still needs correction or explicit exclusion.

Resolving one defect does not resolve a separate impossible measurement.

Changed-case reasoning: Remove that **duplicate** with a recorded reason; C still needs correction or explicit exclusion. Resolving one defect does not resolve a separate impossible measurement.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Keep each learner explaining

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

▼ Earlier wording, alternatives and picture transcripts

Compact alternative: this two-session sequence covers selected core tasks. Use the three-session weekly plan for the full lesson, independent transfer, feedback and project work; allow extra time for extensions.

Demonstrate and guide practice

Board sequence

Write the given facts: Lengths include 5 cm, blank and -3 cm, plus a **duplicate** photo row.

Write the question to explain: Name checks and a deletion risk.

Reveal after **prediction**s: Check **missingness**, physical validity, units and **duplicate** identity. Investigate and document repair. Blind deletion can disproportionately remove a relevant condition.

Use the earlier transfer question as additional practice. Reserve the fresh six-row indoor export for the first individual attempt and record support used.

Classroom dialogue

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

Model these guided questions

Guided: 2 of 5 raw rows have unknown duration, or 40 percent. No: the extra B occurrence is already among the missing rows, so subtracting all three flag counts double-counts it.

A distinct usable-trip count requires a stated resolution policy and source review. Exit examples include **duplicate** IDs, impossible values, units, **provenance** and coverage.

A defensible statement is that this five-row fictional export has documented quality flags and is suitable for inspection practice, with no model-readiness claim.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start to 7 minutes	Retrieve regression target meaning and why later outcomes are separate from prediction inputs. Establish that this week inspects data before fitting.

Stage	Teacher action
7 to 18 minutes	Read the source description and schema. Count physical rows and distinct trip IDs separately. Ask whether a repeated ID proves the reason for duplication.
18 to 32 minutes	Trace the code with row cards. Explain set, None and negative checks. Complete W1 and W2 with named denominators.
32 to 45 minutes	Use the guided question to show overlap: place both B cards in missing and repeated-ID groups. Do not subtract all flag counts independently.
Session 2 start to 8 minutes	Retrieve raw version versus corrected version. Ask what evidence could justify changing a recorded value.
8 to 20 minutes	Read the card example and complete W3. Distinguish a documented fact, an unknown and an assumption.
20 to 35 minutes	Collect W4 independently. Require counts, overlap and a concise card with source and limitations before offering hints.
35 to 45 minutes	Discuss W5 and the exit claim. Use the fresh retry after support and record the support given.

Teach the connected idea

Stage	Teacher action
Session 1 start to 7 minutes	Retrieve the prerequisite and explain the purpose in the opening section.
7 to 20 minutes	Read the worked example and diagram; ask learners to trace each step aloud.
20 to 33 minutes	Complete W1 and W2, then compare predictions with observed results.
33 to 45 minutes	Collect the guided prediction before running; discuss a mismatch and its cause.
Session 2 start to 8 minutes	Retrieve the main rule without showing the previous answer.
8 to 20 minutes	Use W3 to diagnose and repair the stated failure; rerun an earlier case.
20 to 35 minutes	Collect W4 independently, including code or plan, predictions and explanations, before hints.

Stage	Teacher action
35 to 45 minutes	Discuss W5 and the exit; use the fresh retry when support was needed.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

▼ Extension teaching cycles

Connected teaching cycle: Units conversions and data provenance

Target: Convert compatible units before comparing

Reserve further sessions for this topic; the 135-minute primary plan does not absorb all added content.

Completion practice: Convert 150 seconds and 30 seconds to minutes without rounding to whole numbers.

Teacher check: 2.5 and 0.5 minutes.

Individual check: For Units **conversion**s and data **provenance**, use its worked case to evaluate this proposed explanation: “A successful run proves the conclusion”. Explain the deciding step.

Expected correction: Correct **conversion** does not establish that the sources measured the same activity, used the same start and stop rules, or are accurate. **Provenance** helps someone investigate those differences.

Apply the same support, independent transfer and delayed-review decisions to the named connected-topic workbook tasks.

Run the complete supplied investigation

> Keep predictions, observed or supplied outcomes and unmeasured cases separate.

Supplied inputs and checked guidance

Main key: 5 raw/4 IDs/2 missing/1 negative/1 extra ID occurrence; missing and repeated-ID flags overlap on B. Do not infer a usable-trip count by subtraction. Pictured heights have no missing rows; confirmed B copy removal leaves4 rows/4IDs with C still invalid. Probe zero passes, unknown stays unknown,120s converts to2min under known rule,-2 stays flagged.

Independent key6,5,2,1,1; both Q rows overlap missingness with repeated identity. T zero allowed; card states fictional indoor source, intended single trip, minutes, raw unchanged version, findings, and no demonstrated real/outdoor use.

Predict, investigate and record

1. 1. Write the hypothesis and expected **evidence** before running.
2. 2. Data Triage: calculate missing counts, flag **range**s, find **duplicate**s, and write a keep/repair/exclude reason for each issue.
3. 3. Run one normal, one boundary, one missing, and one shifted case.
4. 4. Calculate the metric from raw counts or row-level errors.
5. 5. Interpret what changed, what did not, and what remains unsafe or unknown.

Keep the experiment ledger

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors

Record	What to capture
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose the current misconceptions

A complete card certifies data quality: The card records **evidence** and gaps; its claims still need checking. Documentation does not prove suitability.

Missing means zero: None is unknown; zero is a specific allowed value in this exercise. Substitution without **evidence** changes the meaning.

Flag counts can always be added: The same row can have multiple flags. Count rows or unique events under an explicit policy and retain the raw denominator.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.

What the response shows	Next teaching action
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ Extension misconceptions

A successful run proves the conclusion: Correct **conversion** does not establish that the sources measured the same activity, used the same start and stop rules, or are accurate.

Provenance helps someone investigate those differences.

Matching the example is enough: Require the new W4 case, explicit **prediction**s and an explanation. Fresh retry: 90 seconds, 2 minutes and 30 seconds become 1.5, 2, 0.5; total 4 minutes. Keep the three source records.

A convenient shortcut preserves meaning: Comparing raw 180 to raw 4 ignores their units. Dividing 4 by 60 without checking its unit is also wrong: it was already minutes.

Branch on the documented unit and keep a **conversion** note.

▼ Earlier wording, alternatives and picture transcripts

Run the weekly evidence lab

Keep the test conditions fixed, record every result, and change only one planned factor.

Make the lab runnable

Use the supplied complete investigation for shared practice. Reserve the fresh six-row indoor export for the individual first attempt before hints or feedback; earlier transfer questions remain additional practice.

Diagnose and reteach

MISCONCEPTION TO REPAIR A model will automatically ignore harmless data-quality problems.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

Check these misconceptions

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the selected experiment transfer

> Keep this key until after a first attempt; record any help and prior exposure.

Selected independent task and checked key

Selected case: A new fictional indoor trip export has (P,4), (Q,None), (Q,None), (R,-1), (S,6), (T,0), with durations in minutes and the same **schema**.

Count raw rows, distinct IDs, missing durations, negative durations and extra ID occurrences. Identify overlap. Draft a short **dataset card** stating source, row meaning, unit, findings, unchanged version and one excluded use.

There are 6 rows, 5 distinct IDs, 2 missing durations, 1 negative duration and 1 extra ID occurrence. Both Q rows are missing, so the **duplicate** and missing flags overlap. Zero for T is allowed by the stated **schema**.

The card should identify the supplied fictional indoor export, one intended row per trip, minutes, all findings, raw unchanged version and no **evidence** for real or outdoor navigation use.

Judge trace, scope and support

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one repair action

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Define the row and source

For Robot practice v1, state what one row is intended to represent, the source, the unit and the meaning of None.

One row is intended to represent one indoor practice trip. The source is a teacher-authored fictional export for this lesson, duration uses minutes, and None means unknown rather than zero.

W2 Interpret the audit counts

Explain each number in 5 4 2 1. How many extra ID occurrences are there, and how many rows belong to a repeated-ID group?

The numbers mean 5 raw rows, 4 distinct IDs, 2 missing durations and 1 negative duration. There is 1 extra ID occurrence; 2 rows belong to the repeated-ID group B. These are different counts.

W3 Reject an unsupported correction

A learner changes None to 0 and -2 to 2, then reports a clean dataset. Explain why this is not a justified repair and what record should be kept.

Those edits invent outcomes without source **evidence**. Preserve the raw version, flag the affected rows and investigate.

Record any supported correction with old value, new value, reason and version; unresolved values remain uncertain.

W5 State an evidence gap

Someone **duplicate**s each row ten times and claims the export now covers more kinds of trips. Explain what changed and what did not.

The row count increased but no new trip conditions or independent **observation**s were added. Coverage remains limited to the same supplied fictional indoor cases; the larger count does not establish broader usefulness.

▼ Earlier wording, alternatives and picture transcripts

Assess and explain the answers

Worked case answer

Check **missingness**, physical validity, units and **duplicate** identity. Investigate and document repair. Blind deletion can disproportionately remove a relevant condition.

Additional transfer practice question

A table has height_cm values 12, 15, -8 and a blank. Identify two issues and a suitable first action.

Reasoned practice answer

The negative height is implausible for this measurement and the blank is missing. Check source records and units before choosing a documented correction or exclusion.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “Replace every suspicious value with zero without investigating.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Check -2 against source and units because physical length is nonnegative; mark missing explicitly; identify the **duplicate** and preserve **provenance**; keep 8 if valid.

Document repairs and avoid automatically deleting conditions disproportionately represented among problematic rows.

Explanations for the additional workbook examples

W1 Define the row and source

W2 Interpret the audit counts

W3 Reject an unsupported correction

W4 Inspect and document independently

A new fictional indoor trip export has (P,4), (Q,None), (Q,None), (R,-1), (S,6), (T,0), with durations in minutes and the same **schema**.

Count raw rows, distinct IDs, missing durations, negative durations and extra ID occurrences. Identify overlap. Draft a short **dataset card** stating source, row meaning, unit, findings, unchanged version and one excluded use.

There are 6 rows, 5 distinct IDs, 2 missing durations, 1 negative duration and 1 extra ID occurrence. Both Q rows are missing, so the **duplicate** and missing flags overlap. Zero for T is allowed by the stated **schema**.

The card should identify the supplied fictional indoor export, one intended row per trip, minutes, all findings, raw unchanged version and no **evidence** for real or outdoor navigation use.

W5 State an evidence gap

Assess independent transfer

Selected case answer

Feedback that the learner uses

▼ **Extension questions and checked keys**

Units conversions and data provenance — W1 Retain the source

Name four pieces of information to keep for Log A and its **conversion**.

Source A, raw 180, unit seconds and derived 3.0 minutes; also record divide by 60.

Units conversions and data provenance — W2 Convert precisely

Convert 150 seconds and 30 seconds to minutes without rounding to whole numbers.

2.5 and 0.5 minutes.

There are 60 seconds in one minute. Divide each seconds value by 60: 150/60 = 2.5 minutes and 30/60 = 0.5 minutes. Do not round away the half-minute. Check the direction by converting back: 2.5 times 60 is 150.

Units conversions and data provenance — W3 Repair a comparison

A report claims 120 seconds exceeds 3 minutes because 120 > 3. Correct it.

120 seconds is 2 minutes, so 3 minutes is longer.

First put both values in the same unit. 120 seconds divided by 60 is 2 minutes. Now compare 2 minutes with 3 minutes: 3 minutes is longer by 1 minute.

Comparing the raw numbers 120 and 3 without their units compares unlike quantities.

Units conversions and data provenance — W4 Normalize independently

Create records for 240 seconds, 1.5 minutes and 60 seconds. Derive minutes while keeping raw data. State the values and their total.

4.0, 1.5 and 1.0 minutes; total 6.5 minutes. Retain original units and values.

Units conversions and data provenance — W5 Handle an unknown unit

A record contains value 8 and unit ticks with no **conversion** rule. Should you assume seconds?

No. Mark it unresolved and ask for the unit definition. Do not include it as a known duration.

MAIN LESSON · SECTION 6

Connect, support and revisit the experiment

> Retain failures and assess a fresh parallel case after help.

Project evidence

Audit the Plant Classifier table and attach a data-quality report before any baseline or model is trained.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Support and selected reassessment

Use a tiny labelled dataset, explicit units and a named positive class. Stretch: require a baseline, held-out evaluation and subgroup or shifted-condition analysis with denominators.

Use physical row cards and overlapping circles for flags. Supply blank card headings but no completed findings.

Extension: explain how to record an **evidence**d seconds-to-minutes **conversion** while preserving the **raw value**, unit and version, and why a unit guess is not enough.

Require W4 counts 6,5,2,1,1 in the stated order, the Q overlap, acceptance of zero under this **schema**, and a card with source, unit, raw version and excluded use. Fresh retry: rows (L,3), (M,None), (N,-2), (N,-2), (O,0).

Counts are 5 raw, 4 IDs, 1 missing, 2 negative and 1 extra ID occurrence. The negative and repeated-ID flags overlap on N. Do not infer whether the two N rows describe one event without checking the source.

Delayed return

Suggested return points in this level: ai-4/6, ai-4/8, ai-4/11. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	

Record	Teacher evidence
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Bridge and reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Next: Week 6: Label Quality. Retrieve today's **evidence** and identify what carries forward and what is new.

Next week is Week 6: Label Quality. Carry forward the **dataset card**, source questions and explicit distinction between unknown and observed values. Teacher reference: https://arxiv.org/abs/1803.09010 (Datasheets for Datasets).

▼ Earlier wording, alternatives and picture transcripts

Connect and extend learning

Responsible Plant Classifier

Support and stretch

Exit reflection

Bridge to the next week

Support and fresh reassessment

Plan the delayed return

Project evidence from this case

▼ **Extension support and reassessment**

Readiness and reassessment for Units conversions and data provenance

Use 60-second strips. Extension: compare results before and after rounding and explain why early rounding can change totals.

Independent W4 **evidence**: 4.0, 1.5 and 1.0 minutes; total 6.5 minutes. Retain original units and values. Record whether the learner needed no help, a hint or explanation.

Fresh retry: 90 seconds, 2 minutes and 30 seconds become 1.5, 2, 0.5; total 4 minutes. Keep the three source records.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route without repeating the whole cycle.

Planning and assessment reference

Teaching target: Inspect row meaning **missing value**s and repeated records

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

Before class, solve the supported practice: Explain each number in 5 4 2 1. How many extra ID occurrences are there, and how many rows belong to a repeated-ID group?

Reasoned practice answer: The numbers mean 5 raw rows, 4 distinct IDs, 2 missing durations and 1 negative duration. There is 1 extra ID occurrence; 2 rows belong to the repeated-ID group B. These are different counts.

Check every learner before discussion: Use the worked case for Dataset inspection and cards. Proposed explanation: “A complete card certifies data quality”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: The card records **evidence** and gaps; its claims still need checking. Documentation does not prove suitability.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: The teacher confirms B’s second row is an accidental **duplicate**. Require a **prediction** and a reason before comparing.

Expected result: Remove that **duplicate** with a recorded reason; C still needs correction or explicit exclusion. Reason: Resolving one defect does not resolve a separate impossible measurement.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback and evidence records**

REFERENCE · SECTION 8

Reference · Feedback and evidence records

> Keep supported rehearsal separate from independent performance.

Record and repair

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.
</details>

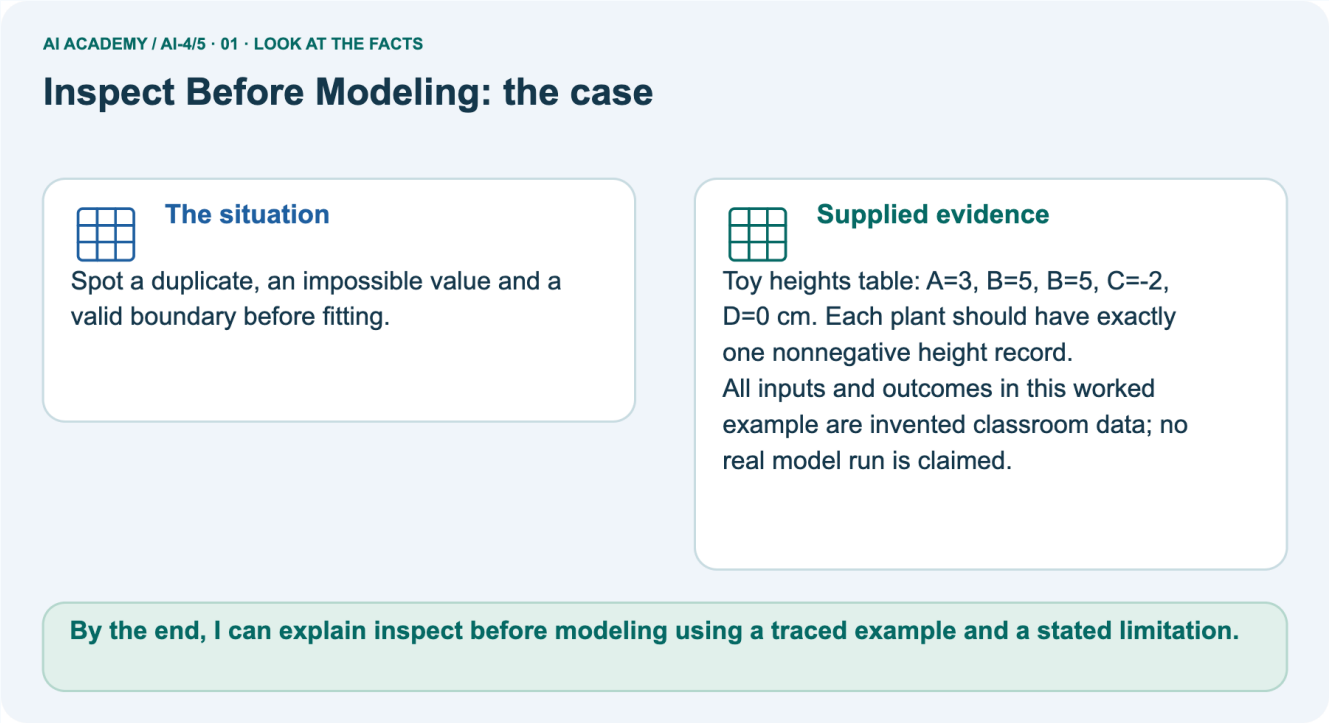
<details>
<summary>Reference · Teach with the six pictures</summary>
</details>

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a specific reasoning move.

01. Inspect Before Modeling: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

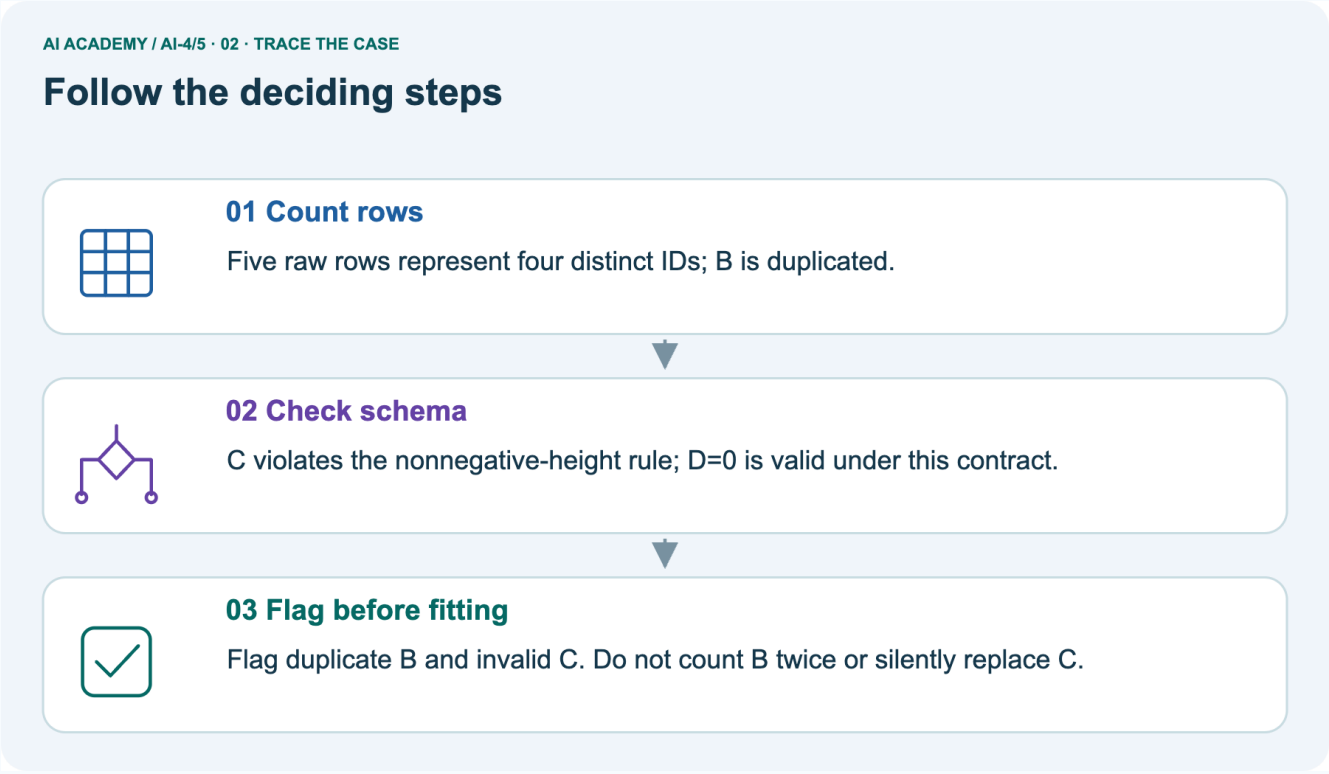
Spot a **duplicate**, an impossible value and a valid boundary before **fitting**. Supplied facts: Toy heights table: A=3, B=5, B=5, C=-2, D=0 cm. Each plant should have exactly one nonnegative height record.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

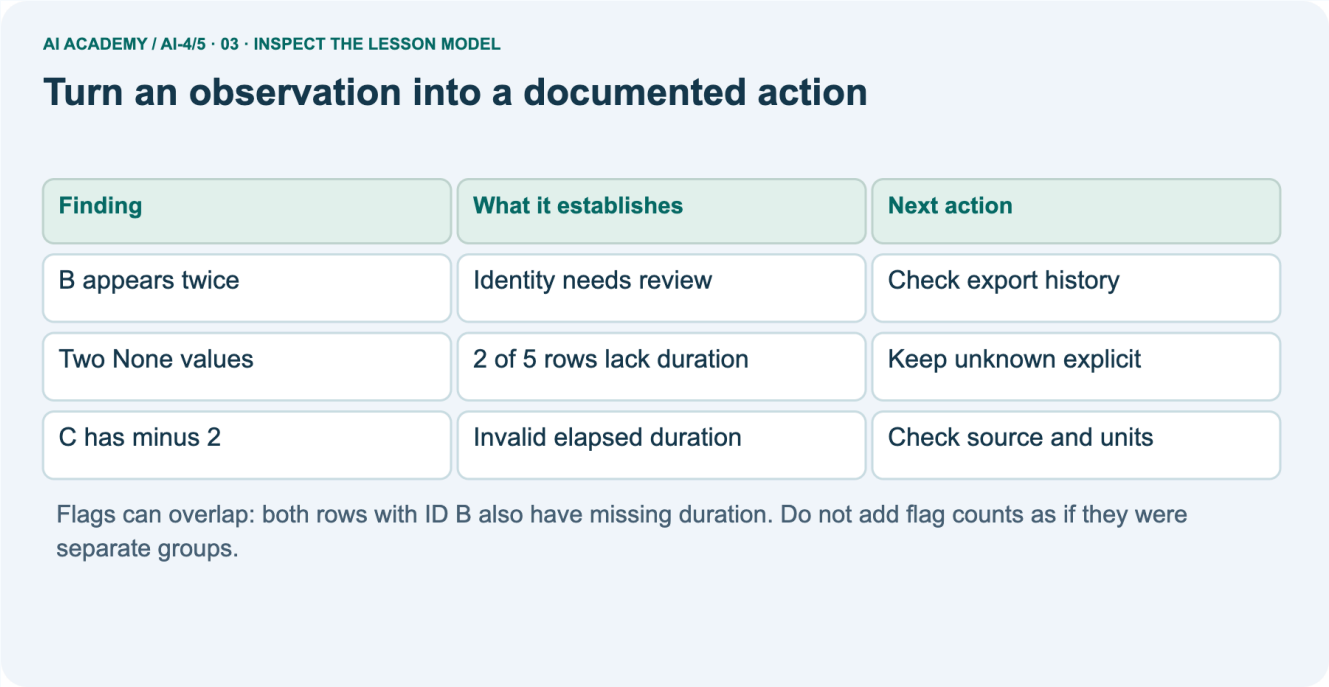
Count rows: Five raw rows represent four distinct IDs; B is **duplicate**d. → Check **schema**: C violates the nonnegative-height rule; D=0 is valid under this contract. → Flag before **fitting**: Flag **duplicate** B and invalid C.

Do not count B twice or silently replace C.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Turn an observation into a documented action



Follow the labelled picture with the matching explanation. Open the vector diagram

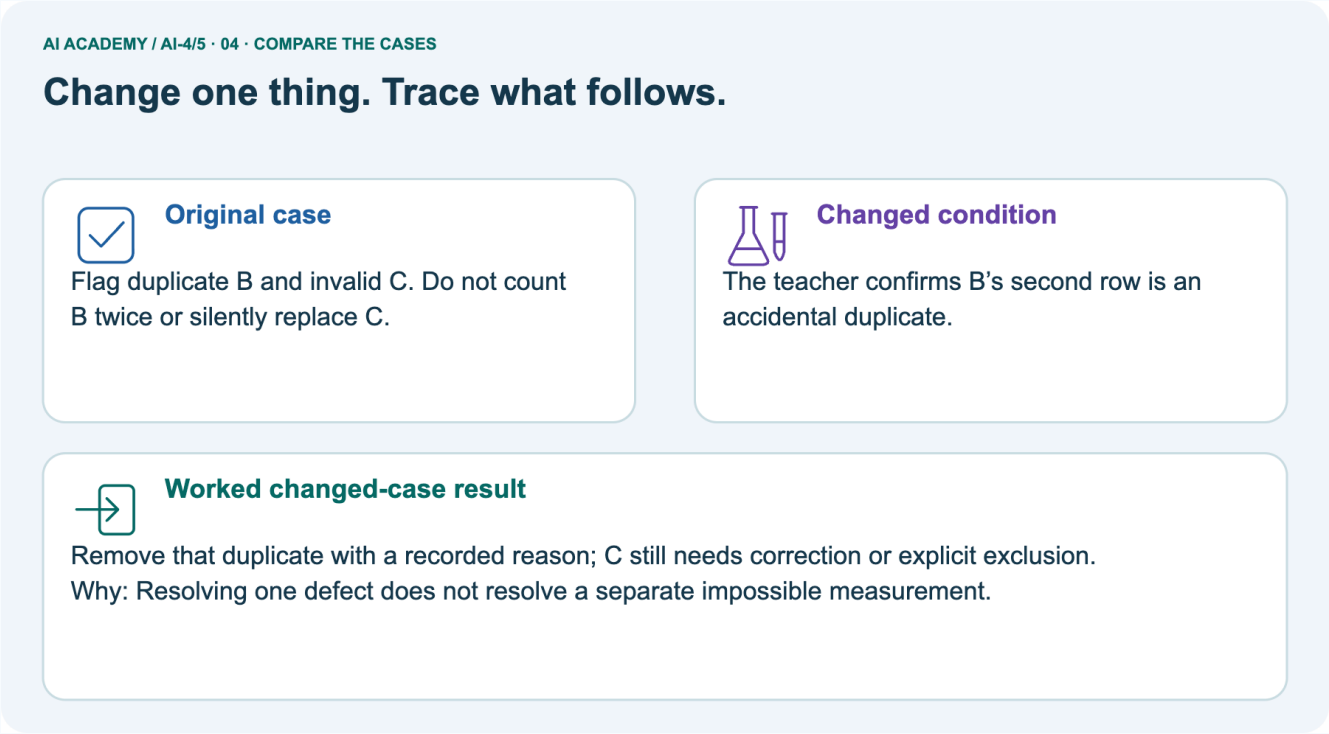
Turn an **observation** into a documented action. B appears twice | Identity needs review | Check export history. Two None values | 2 of 5 rows lack duration | Keep unknown explicit.

C has minus 2 | Invalid elapsed duration | Check source and units

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Flag **duplicate** B and invalid C. Do not count B twice or silently replace C. Changed condition: The teacher confirms B's second row is an accidental **duplicate**.

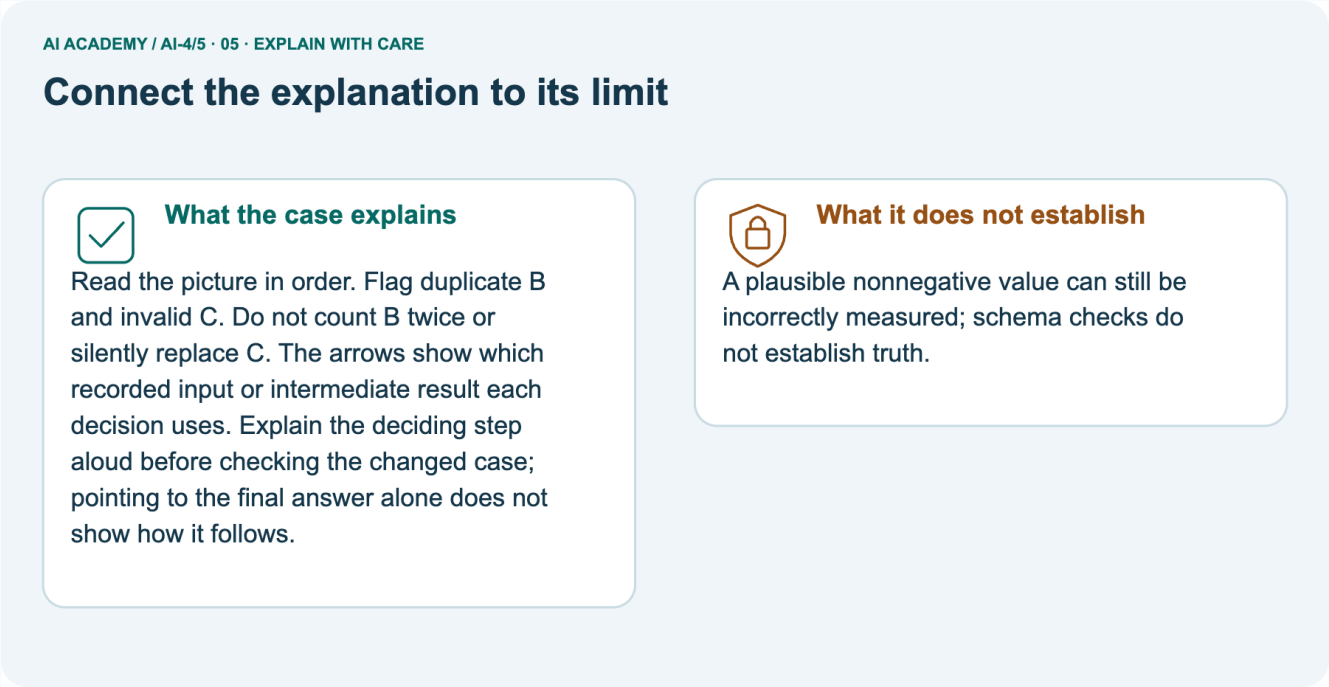
Expected result: Remove that **duplicate** with a recorded reason; C still needs correction or explicit exclusion. Reason: Resolving one defect does not resolve a separate impossible measurement.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Resolving one defect does not resolve a separate impossible measurement.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Flag **duplicate** B and invalid C. Do not count B twice or silently replace C. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

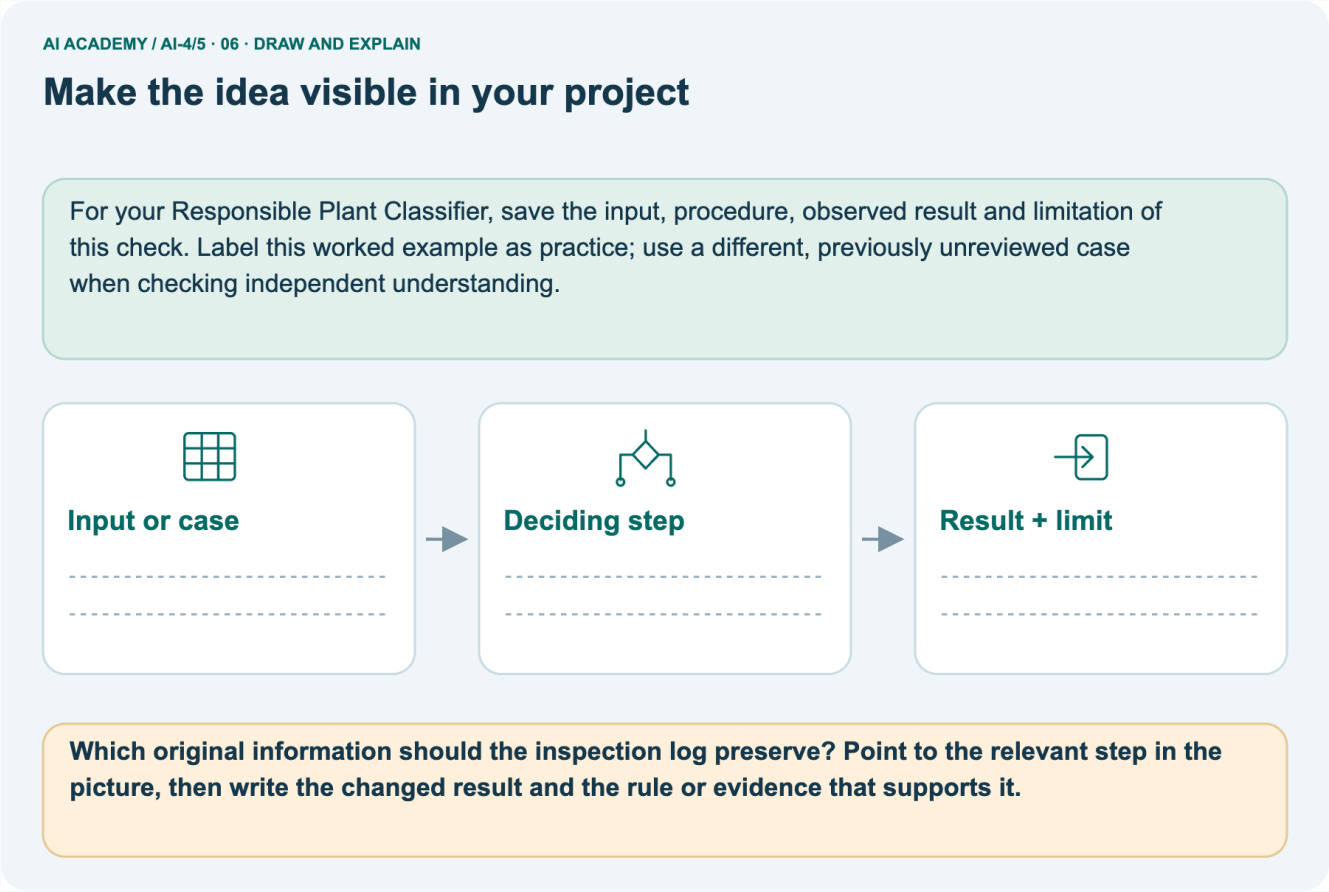
Limit: A plausible nonnegative value can still be incorrectly measured; **schema** checks do not establish truth.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: A plausible nonnegative value can still be incorrectly measured; **schema** checks do not establish truth.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Which original information should the inspection log preserve? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: Keep raw row count 5, distinct IDs 4, **duplicate** B and invalid C, plus any confirmed correction and its reason.

Changed-case check: Remove that **duplicate** with a recorded reason; C still needs correction or explicit exclusion. Resolving one defect does not resolve a separate impossible measurement.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Inspect Before Modeling: the case

02. Follow the deciding steps

03. Turn an observation into a documented action

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.